

Convex Relaxation for Robust Vanishing Point Estimation in Manhattan World

Bangyan Liao^{1,2*}, Zhenjun Zhao^{3*}, Haoang Li⁴, Yi Zhou⁵, Yingping Zeng⁵, Hao Li², Peidong Liu^{2✉}

¹Zhejiang University ²Westlake University

³The Chinese University of Hong Kong

⁴Hong Kong University of Science and Technology (Guangzhou)

⁵Hunan University

Abstract

Determining the vanishing points (VPs) in a Manhattan world, as a fundamental task in many 3D vision applications, consists of jointly inferring the line-VP association and locating each VP. Existing methods are, however, either sub-optimal solvers or pursuing global optimality at a significant cost of computing time. In contrast to prior works, we introduce convex relaxation techniques to solve this task for the first time. Specifically, we employ a “soft” association scheme, realized via a truncated multi-selection error, that allows for joint estimation of VPs’ locations and line-VP associations. This approach leads to a primal problem that can be reformulated into a quadratically constrained quadratic programming (QCQP) problem, which is then relaxed into a convex semidefinite programming (SDP) problem. To solve this SDP problem efficiently, we present a globally optimal outlier-robust iterative solver (called **GlobustVP**), which independently searches for one VP and its associated lines in each iteration, treating other lines as outliers. After each independent update of all VPs, the mutual orthogonality between the three VPs in a Manhattan world is reinforced via local refinement. Extensive experiments on both synthetic and real-world data demonstrate that **GlobustVP** achieves a favorable balance between efficiency, robustness, and global optimality compared to previous works. The code is publicly available at github.com/wu-cvgl/GlobustVP.

1. Introduction

Multiple parallel 3D lines project onto the camera image plane, converging at a specific point known as the vanishing point (VP). Vanishing point estimation is a fundamental task in Simultaneous Localization and Mapping

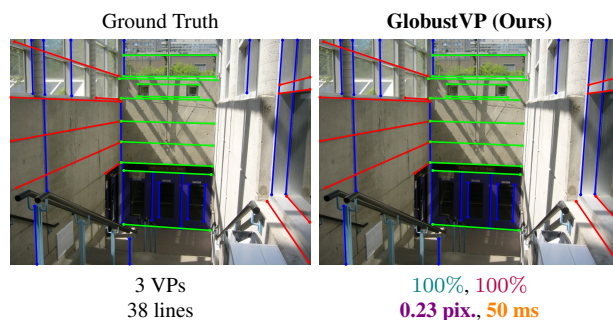


Figure 1. Representative results of our method (**GlobustVP**) on the York Urban Database [10] for estimating three vanishing points (VPs) in the Manhattan world. Different line-VP associations are shown in respective colors. Below each image, we report the widely-used evaluation metrics, **precision** $\uparrow$, **recall** $\uparrow$, **consistency error** $\downarrow$ (defined in Sec. 5.1.1), and **runtime** $\downarrow$. **GlobustVP** achieves high precision and fast runtime, demonstrating an excellent balance between accuracy and efficiency.

(SLAM) [20, 22], camera calibration [19, 23], and structural understanding [12]. In this paper, we focus on vanishing point estimation in the Manhattan world [36], which assumes only three mutually orthogonal line directions and represents the most common structural pattern in man-made environments. The challenge lies in the coupling relationship between line association and vanishing point estimation, which leads to numerous local minima. This characteristic makes an algorithm with a global optimality guarantee particularly valuable.

Previous methods for accurate and efficient vanishing point estimation rely on local, global, or learning-based approaches. Local methods (e.g., Expectation-Maximization [10], Hough voting [40], and RANSAC [11]) require either a good VP initialization or exhaustive sampling to achieve the desired accuracy. Due to the stochastic nature of local solvers, they cannot guarantee global optimality. In contrast, global methods (e.g., Branch-and-Bound [6, 24]) always guarantee the globally optimal solu-

* Equal contribution.

✉ Corresponding author: Peidong Liu (liupeidong@westlake.edu.cn).

tion. However, the exponential time cost in the worst case makes them inefficient, and the large search space hinders their use in large-scale scenarios. Recently, the success of deep learning has led researchers to apply it to vanishing point estimation [27, 39]. However, learning-based methods lack optimality guarantees and often struggle to generalize beyond their training datasets, limiting their practical utility. In conclusion, a globally optimal, outlier-robust, and sufficiently fast vanishing point estimation algorithm remains missing.

To address this, we introduce convex relaxation techniques for the first time in the context of vanishing point estimation. Specifically, we propose a novel truncated multi-selection error formulation that enables joint optimization of three vanishing points and their corresponding line association labels (*e.g.*, VP1, VP2, VP3, Outlier). This error formulation is incorporated into **Primal Problem**, reformulated as **Full QCQP Problem**, and subsequently relaxed into the convex **Full SDP Problem**. To solve such a problem efficiently, we propose an iterative solver **GlobustVP**, which solves one **Single Block SDP Problem** in each iteration to identify a vanishing point and its corresponding inlier line set, which is then excluded from subsequent iterations. After finding all three vanishing points, **Manhattan Post-Refinement** is applied to ensure that the estimated vanishing points satisfy the Manhattan world assumption.

In summary, our **contributions** are as follows:

- A “soft” association scheme, realized via the proposed truncated multi-selection error, for joint estimation of VPs’ location and line-VP association.
- Introducing convex relaxation to reformulate the intermediate QCQP form of the primal problem as a convex SDP problem.
- An iterative solver, **GlobustVP**, that solves each VP sub-problem independently (corresponding to a sub-block of the full SDP problem), achieving global optimality under mild conditions.
- Extensive evaluations on both synthetic and real data demonstrate that our method achieves superior accuracy and robustness, while being on par with prior methods in terms of efficiency.

2. Related Work

Existing vanishing point estimation methods can be broadly categorized into five main classes: Expectation-Maximization (EM)-based [2, 9, 10], Hough-voting-based [4, 13, 29, 31, 33, 40], RANSAC-based [5, 32, 35, 37, 38, 42, 46], Branch-and-Bound (BnB)-based [6, 7, 24, 25], and learning-based methods [17, 26, 27, 39, 41, 44].

EM-based methods [2, 9, 10] iteratively associate image lines to estimate vanishing points, but they are prone to local minima and sensitive to initialization. Hough-voting-based approaches [4, 13, 33] partition the Gaussian sphere

and accumulate votes along the locus of each great circle to identify the dominant directions. However, they are susceptible to bin quantization issues and may detect spurious vanishing points. RANSAC-based methods [38, 46] typically sample subsets of image lines to hypothesize candidate vanishing points and evaluate each hypothesis based on inlier consensus. While achieving robustness against outliers, these approaches are inherently non-deterministic and do not guarantee global optimality due to sampling uncertainty and failure to satisfy orthogonality constraints [32, 37]. BnB-based approaches [6, 24, 25] treat vanishing point estimation as a global optimization problem and use strategies such as branch-and-bound [15, 21] to improve performance. Bazin *et al.* [6] formulate the task as a consensus set maximization problem over the rotation search space and further solve it efficiently using a BnB scheme. While these methods guarantee global optimality, they can be computationally expensive due to the exploratory nature and exponential worst-case performance of BnB. Hybrid approaches [24, 25] combine BnB and RANSAC to improve efficiency, but only achieve quasi-global optimality.

Recently, learning-based methods have demonstrated effectiveness in vanishing point estimation. Zhai *et al.* [41] use a deep convolutional neural network (CNN) to extract the global image context and generate horizon line candidates. NeurVPS [44] employs conic convolutions to refine feature extraction along structural lines. Lin *et al.* [27] incorporate the Hough Transform and Gaussian sphere into a deep vanishing point detection network. More recently, methods like CONSAC [17] and PARSAC [16] combine neural networks with multi-model fitting to improve vanishing point estimation. While these methods achieve promising results, they often involve trade-offs between computational efficiency and generalization, particularly when compared to traditional RANSAC- or BnB-based approaches.

In summary, achieving a balance between computational efficiency, robustness to outliers, and global optimality remains a key challenge in vanishing point estimation. Our approach addresses these limitations by incorporating a convex relaxation technique, offering a favorable trade-off between efficiency, robustness, and optimality. Extensive experiments further demonstrate the effectiveness of our method across a variety of settings.

3. Notations and Background

3.1. Notation

We use lowercase letters (*e.g.*, s) to denote scalars, bold lowercase letters (*e.g.*, $\mathbf{v}$) for vectors, and bold uppercase letters (*e.g.*, $\mathbf{M}$) for matrices. $[\mathbf{M}]_{i,j}$ denotes the scalar entry in the i^{th} row and j^{th} column of the matrix $\mathbf{M} \in \mathbb{R}^{m \times n}$. $\{\mathbf{M}\}_{i,j}$ denotes the block sub-matrix at the i^{th} row and j^{th} column of the matrix $\mathbf{M}$. $[\mathbf{M}]_{i,*}$ denotes the i^{th} row of the

matrix $\mathbf{M}$. The operation $(\mathbf{M})^2 : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$ denotes the element-wise square operator. $\mathbf{I}_d$ is the identity matrix of size d . The vectors $\mathbf{1}_n \in \mathbb{R}^n$ and $\mathbf{0}_n \in \mathbb{R}^n$ denote all-one and all-zero vectors, respectively. The $[\mathbf{M}; \mathbf{N}]$ operator denotes the matrix stacking operator, while the $[\mathbf{M}, \mathbf{N}]$ operator denotes matrix concatenation. We use “ $\otimes$ ” to denote the Kronecker product. $\text{vec}(\mathbf{M})$ denotes the vectorization operator, which concatenates the columns of $\mathbf{M}$. $\text{diag}([\mathbf{M}_1, \dots, \mathbf{M}_n])$ denotes the operation of arranging the square matrices $\mathbf{M}_1$ to $\mathbf{M}_n$ along the diagonal.

3.2. Geometry Related to Vanishing Point

The vanishing point is a widely used concept in 3D vision, based on the observation that two parallel lines in a Euclidean space intersect at a point in the image projection space, known as the vanishing point [14, 42]. This constraint can be expressed as:

$$\mathbf{v}^\top \mathbf{l} = 0, \quad (1)$$

where $\mathbf{v} \in \mathbb{R}^3$ and $\mathbf{l} \in \mathbb{R}^3$ denote the homogeneous vanishing point and its associated line on image plane, respectively. We can convert $\mathbf{v}$ and $\mathbf{l}$ to the normalized image coordinate system:

$$\mathbf{d}^\top \mathbf{n} = 0, \quad (2)$$

$$\mathbf{v} = \mathbf{K}\mathbf{d}, \quad \mathbf{l} = \mathbf{K}^{-\top} \mathbf{n}, \quad (3)$$

where $\mathbf{K} \in \mathbb{R}^{3 \times 3}$ is the camera intrinsic matrix [14], $\mathbf{d} \in \mathbb{R}^3$ represents the 3D line direction (equivalent to the vanishing point), and $\mathbf{n} \in \mathbb{R}^3$ is the normal vector to the projection plane for each image line. For simplicity, we will adopt the normalized coordinate system in the following derivations. Additionally, the Manhattan world assumption implies that the directions of the three 3D lines $\mathbf{d}$ are mutually orthogonal, forming a 3×3 orthogonal matrix. Leveraging this Manhattan world prior constraint can enhance the stability of the overall optimization algorithm.

3.3. Convex Relaxation

Given a general Quadratically Constrained Quadratic Programming (QCQP) formulation:

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^n} \quad & \mathbf{x}^\top \mathbf{C} \mathbf{x} \\ \text{s.t.} \quad & \mathbf{x}^\top \mathbf{A}_i \mathbf{x} = b_i, \quad i = 1, \dots, m, \end{aligned} \quad (4)$$

where $\mathbf{C}, \mathbf{A}_1, \dots, \mathbf{A}_m \in \mathbb{S}^n$ are real symmetric matrices. Using the matrix trace property, we can convert the quadratic error term as:

$$\mathbf{x}^\top \mathbf{C} \mathbf{x} = \text{trace}(\mathbf{x}^\top \mathbf{C} \mathbf{x}) = \text{trace}(\mathbf{C} \mathbf{x} \mathbf{x}^\top). \quad (5)$$

Thus, by introducing a new rank-one symmetric positive semidefinite (PSD) matrix $\mathbf{X} = \mathbf{x} \mathbf{x}^\top$, we can reformulate

the above QCQP as:

$$\begin{aligned} \min_{\mathbf{X} \in \mathcal{S}^n} \quad & \text{trace}(\mathbf{C} \mathbf{X}) \\ \text{s.t.} \quad & \text{trace}(\mathbf{A}_i \mathbf{X}) = b_i, \quad i = 1, \dots, m, \\ & \mathbf{X} \succeq \mathbf{0}, \quad \text{rank}(\mathbf{X}) = 1. \end{aligned} \quad (6)$$

By relaxing the rank constraint, we obtain the standard semidefinite programming (SDP) formulation:

$$\begin{aligned} \min_{\mathbf{X} \in \mathcal{S}^n} \quad & \text{trace}(\mathbf{C} \mathbf{X}) \\ \text{s.t.} \quad & \text{trace}(\mathbf{A}_i \mathbf{X}) = b_i, \quad i = 1, \dots, m, \\ & \mathbf{X} \succeq \mathbf{0}. \end{aligned} \quad (7)$$

The primary benefit of employing the convex relaxation, rather than solving the original problem, lies in the convex nature of the resulting formulation. Although relaxation has been applied, researchers have found that strong duality properties still hold for many problems [8, 43]. This implies that solving the relaxed problem is equivalent to solving the original one [1]. Leveraging any off-the-shelf SDP solver [3], we can always find the global optimum in polynomial time.

4. Methodology

In Sec. 4.1, we formulate the vanishing point estimation and line association problem as **Primal Problem** using the proposed truncated multi-selection error. In Sec. 4.2, we reformulate this **Primal Problem** as **Full QCQP Problem** and subsequently rewrite it as **Full SDP Problem**. To solve such a SDP problem efficiently, we introduce an iterative solver **GlobustVP** in Sec. 4.3, which solves one instance of **Single Block SDP Problem** per iteration, followed by **Manhattan Post-Refinement** after finding all three VPs.

4.1. Truncated Multi-selection Error Formulation

We begin by defining the VP matrix $\mathbf{D} = [\mathbf{d}_1^\top; \mathbf{d}_2^\top; \mathbf{d}_3^\top] \in \mathbb{R}^{3 \times 3}$, where each $\mathbf{d}_i \in \mathbb{R}^3$ represents a 3D line direction (equivalent to the vanishing point). Next, we define the line matrix $\mathbf{N} = [\mathbf{n}_1, \mathbf{n}_2, \dots, \mathbf{n}_m] \in \mathbb{R}^{3 \times m}$, where each $\mathbf{n}_i \in \mathbb{R}^3$ represents a unit normal perpendicular to the projection plane (equivalent to an image line), and m denotes the number of lines in the image. By multiplying the VP matrix $\mathbf{D}$ with the line matrix $\mathbf{N}$, stacking it with $c\mathbf{1}_m^\top$, and then performing element-wise squaring, we obtain the distance matrix $([\mathbf{D}\mathbf{N}; c\mathbf{1}_m^\top])^2 \in \mathbb{R}^{4 \times m}$. Each element in $(\mathbf{D}\mathbf{N})^2 \in \mathbb{R}^{3 \times m}$ represents the point-to-line distance between different VPs and lines, as shown in Fig. 2a. Finally, we define the permutation matrix as $\mathbf{Q} \in \mathbb{R}^{4 \times m}$, where each column indicates the association label of each line, as shown in Fig. 2b. In summary, the primal problem is de-

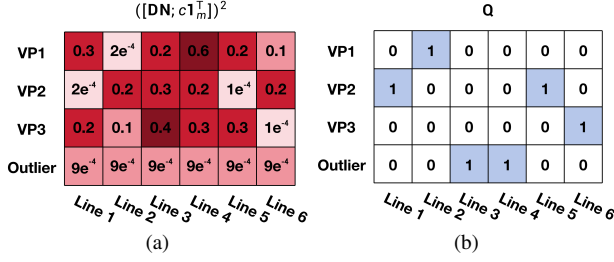


Figure 2. An example of the distance matrix $([DN; c\mathbf{1}_m^T])^2$ and the associated permutation matrix $\mathbf{Q}$. In this example, $c = 0.03$, line 1 matches VP2, line 2 matches VP1, line 3 and 4 are identified as outliers, line 5 matches VP2, and line 6 matches VP3.

fined as:

$$\begin{aligned}
 \min_{\mathbf{D}, \mathbf{Q}} \quad & \langle ([DN; c\mathbf{1}_m^T])^2, \mathbf{Q} \rangle \\
 \text{s.t.} \quad & \mathbf{Q}^T \mathbf{1}_4 = \mathbf{1}_m, (\mathbf{Q})^2 = \mathbf{Q}, \\
 & [\mathbf{D}]_{1,*} [\mathbf{D}]_{1,*}^T = 1, [\mathbf{D}]_{2,*} [\mathbf{D}]_{2,*}^T = 1, \\
 & [\mathbf{D}]_{3,*} [\mathbf{D}]_{3,*}^T = 1, [\mathbf{D}]_{1,*} [\mathbf{D}]_{2,*}^T = 0, \\
 & [\mathbf{D}]_{2,*} [\mathbf{D}]_{3,*}^T = 0, [\mathbf{D}]_{1,*} [\mathbf{D}]_{3,*}^T = 0,
 \end{aligned}$$

(Primal Problem)

where c denotes the maximum inlier threshold.

Remarks. 1) The error formulation is designed to be both truncated and multi-selection. “Truncated” means that outlier lines are assigned a maximum inlier threshold (i.e., c^2). This occurs when the distance between a line and all vanishing points exceeds the maximum inlier threshold (e.g., lines 3 and 4 in Fig. 2). In this case, the corresponding matrix $\mathbf{Q}$ will label the line as an outlier, set its error to zero, and add a penalty term of c^2 . “Multi-selection” means that we can not only label inliers and outliers, but also identify the vanishing point to which each line belongs. 2) The first row of constraints ensures that the elements of the permutation matrix $\mathbf{Q}$ are binary (either “0” or “1”), with exactly single “1” in each column. The remaining constraints enforce the orthogonality of the VP matrix $\mathbf{D}$. 3) It is clear that when the distance between all outlier lines and all ground-truth vanishing points exceeds the maximum inlier threshold (i.e., c^2), the global optimum of [Primal Problem](#) corresponds to the largest set of inliers and the optimal vanishing point estimation. In this case, the error of the outliers is set to a constant value, ensuring that no bias is introduced.

4.2. Full QCQP Formulation and Convex Relaxation

4.2.1. Full QCQP Formulation

Given the primal formulation [Primal Problem](#), we will then derive the equivalent QCQP formulation. To begin with, we firstly introduce a vector $\bar{\mathbf{D}} = [\mathbf{d}_1; \mathbf{d}_2; \mathbf{d}_3; \mathbf{1}] \in \mathbb{R}^{10}$ and a

new variable $\boldsymbol{\omega} \in \mathbb{R}^{10(1+4m)}$:

$$\begin{aligned}
 \boldsymbol{\omega} &= [\bar{\mathbf{D}}; \text{vec}(\mathbf{Q}) \otimes \bar{\mathbf{D}}] \\
 &= [\omega_0; \omega_{1,1}; \dots; \omega_{4,1}; \omega_{1,2}; \dots; \omega_{4,m}],
 \end{aligned}$$

(8)

where

$$\begin{cases} \omega_0 = \bar{\mathbf{D}} \in \mathbb{R}^{10}, & \omega_{i,j} = [\mathbf{Q}]_{i,j} \bar{\mathbf{D}} \in \mathbb{R}^{10}, \\ \{\omega_0\}_i = \mathbf{d}_i \in \mathbb{R}^3 \quad (i = 1, 2, 3). \end{cases}$$

(9)

Additionally, we introduce an auxiliary matrix $\mathbf{C} \in \mathbb{R}^{10(1+4m) \times 10(1+4m)}$, which is defined as a block diagonal matrix:

$$\mathbf{C} = \text{diag}([\mathbf{C}_0, \mathbf{C}_{1,1}, \dots, \mathbf{C}_{4,1}, \mathbf{C}_{1,2}, \dots, \mathbf{C}_{4,m}]),$$

(10)

where

$$\begin{cases} \mathbf{C}_0 = \mathbf{0}_{10 \times 10}, \\ \{\mathbf{C}_{i,j}\}_{i,j} = \mathbf{n}_j \mathbf{n}_j^T, & i = 1, \dots, 3, \quad j = 1, \dots, m, \\ [\mathbf{C}_{4,j}]_{10,10} = c^2, & j = 1, \dots, m. \end{cases}$$

(11)

Finally, we can equivalently transform [Primal Problem](#) into the following QCQP problem:

$$\begin{aligned}
 \min_{\boldsymbol{\omega}} \quad & \boldsymbol{\omega}^T \mathbf{C} \boldsymbol{\omega} \\
 \text{s.t.} \quad & \omega_0 \omega_0^T = \sum_{i=1}^4 \omega_{0,i} \omega_{0,i}^T, \quad j = 1, \dots, m, \\
 & \omega_0 \omega_{i,j}^T = \omega_{i,j} \omega_{i,j}^T, \quad \begin{cases} i = 1, \dots, 4 \\ j = 1, \dots, m \end{cases}, \\
 & \{\omega_0\}_1^T \{\omega_0\}_1 = 1, \quad \{\omega_0\}_2^T \{\omega_0\}_2 = 1, \\
 & \{\omega_0\}_3^T \{\omega_0\}_3 = 1, \quad \{\omega_0\}_1^T \{\omega_0\}_2 = 0, \\
 & \{\omega_0\}_1^T \{\omega_0\}_3 = 0, \quad \{\omega_0\}_2^T \{\omega_0\}_3 = 0.
 \end{aligned}$$

(Full QCQP Problem)

This QCQP formulation will serve as a bridge to derive the subsequent SDP formulation.

Remark. The [Full QCQP Problem](#) is equivalent to the [Primal Problem](#). In general, the first and second constraints are equivalent to $\mathbf{Q}^T \mathbf{1}_4 = \mathbf{1}_m$ and $(\mathbf{Q})^2 = \mathbf{Q}$, respectively. The remaining constraints enforce the orthogonality of the VP matrix $\mathbf{D}$.

4.2.2. Semidefinite Programming Formulation

Given the definition of [Full QCQP Problem](#), we can relax it to the [Full SDP Problem](#) as:

$$\begin{aligned}
 \min_{\mathbf{W}} \quad & \text{trace}(\mathbf{C}\mathbf{W}) \\
 \text{s.t.} \quad & \mathbf{W}_{0,0} = \sum_{i=1}^4 \mathbf{W}_{0,4(j-1)+i}, \quad j = 1, \dots, m, \\
 & \mathbf{W}_{0,k} = \mathbf{W}_{k,k}, \quad k = 1, \dots, 4m, \\
 & \text{trace}(\{\mathbf{W}_{0,0}\}_{i,j}) = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases} \quad \forall i, j \in \{1, 2, 3\} \\
 & \mathbf{W} \succeq \mathbf{0},
 \end{aligned}$$

(Full SDP Problem)

where

$$\mathbf{W} \in \mathcal{S}_+^{10(1+4m) \times 10(1+4m)}$$

$$= \begin{bmatrix} \mathbf{W}_{0,0} & \mathbf{W}_{0,1} & \cdots & \mathbf{W}_{0,4m} \\ \mathbf{W}_{1,0} & \mathbf{W}_{1,1} & \cdots & \mathbf{W}_{1,4m} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{W}_{4m,0} & \mathbf{W}_{4m,1} & \cdots & \mathbf{W}_{4m,4m} \end{bmatrix}. \quad (12)$$

Remark. The convex relaxation strategy transforms [Full QCQP Problem](#) into a convex problem, specifically [Full SDP Problem](#), by discarding the only non-convex rank constraint $\text{rank}(\mathbf{W}) = 1$. The SDP problem can be globally solved using any off-the-shelf solver in polynomial time. Furthermore, we provide a proof of tight relaxation (*i.e.*, zero duality gap) in Supplementary Material Sec. 1.2, which demonstrates that solving [Full SDP Problem](#) is equivalent to solving [Primal Problem](#).

4.3. GlobustVP: Iterative SDP Solver

To solve [Full SDP Problem](#) efficiently, we propose an iterative solver (called **GlobustVP**) to estimate the vanishing points one by one. Specifically, in each iteration, we solve [Single Block SDP Problem](#) (Sec. 4.3.1) to globally search for a single vanishing point, treating the other vanishing points as outliers. In contrast to the previous algorithm, where all three vanishing points were estimated simultaneously, this approach estimates them sequentially. After each iteration, the inlier lines are marked and then excluded from subsequent iterations. After three iterations, [Manhattan Post-Refinement](#) (Sec. 4.3.2) is applied to ensure that the vanishing points satisfy the Manhattan world assumption. The algorithm pipeline is summarized in Alg. 1.

4.3.1. Single Block SDP Problem

We first define the tensor $\mathbf{W} \in \mathbb{R}^{3(m+1) \times 3(m+1) \times 2}$ with the following block structure:

$$\mathbf{W}_{*,*,i} = \begin{bmatrix} \mathbf{W}_{0,0,i} & \mathbf{W}_{0,1,i} & \cdots & \mathbf{W}_{0,m,i} \\ \mathbf{W}_{1,0,i} & \mathbf{W}_{1,1,i} & \cdots & \mathbf{W}_{1,m,i} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{W}_{m,0,i} & \mathbf{W}_{m,1,i} & \cdots & \mathbf{W}_{m,m,i} \end{bmatrix}. \quad (13)$$

Additionally, the auxiliary tensor $\mathbf{C} \in \mathbb{R}^{3(m+1) \times 3(m+1) \times 2}$ is defined as:

$$\begin{cases} \mathbf{C}_{*,*,1} = \text{diag}([\mathbf{0}_{3 \times 3}, \mathbf{n}_1 \mathbf{n}_1^\top, \dots, \mathbf{n}_m \mathbf{n}_m^\top]), \\ \mathbf{C}_{*,*,2} = \text{diag}([\mathbf{0}_{3 \times 3}, c^2 \mathbf{I}_3, \dots, c^2 \mathbf{I}_3]). \end{cases} \quad (14)$$

Finally, we convert [Full SDP Problem](#) to its iterative vari-

Algorithm 1 GlobustVP Solver

Input: A set of detected lines, $\mathbf{n} = \{\mathbf{n}_1, \mathbf{n}_2, \dots, \mathbf{n}_m\}$

Output: Three globally optimal vanishing points and the set of line labels, indicating line associations as VP1, VP2, VP3, Outlier.

- 1: **for** $i = 1 : 3$ **do**
 - 2: Construct the auxiliary tensor $\mathbf{C}_i$ for the line set $\mathbf{n}$ following Eq. (14),
 - 3: Solve the [Single Block SDP Problem](#) to obtain $\mathbf{W}_i$,
 - 4: Round $\mathbf{W}_i$ and recover one vanishing point, $\mathbf{d}_i$,
 - 5: Collect the inlier line IDs in set $\mathbf{l}_i$ and remove these lines from the set $\mathbf{n}$,
 - 6: **end for**
 - 7: Collect the remaining line IDs in set $\mathbf{l}_{\text{outlier}}$,
 - 8: Perform the [Manhattan Post-Refinement](#) initialized with $\{\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3\}$.
-

ant, [Single Block SDP Problem](#), as:

$$\begin{aligned} \min_{\mathbf{W}} \quad & \text{trace}(\mathbf{C}\mathbf{W}) \\ \text{s.t.} \quad & \mathbf{W}_{0,0,1} = \sum_{i=1}^2 \text{trace}(\mathbf{W}_{0,j,i}), \quad j = 1, \dots, m, \\ & \mathbf{W}_{0,j,i} = \mathbf{W}_{j,j,i}, \quad \forall i \in \{1, 2\}, \quad j = 1, \dots, m, \\ & \text{trace}(\mathbf{W}_{0,0,1}) = 1, \quad \mathbf{W}_{0,0,1} = \mathbf{W}_{0,0,2}, \\ & \mathbf{W}_{*,*,i} \succeq \mathbf{0}, \quad \forall i \in \{1, 2\}. \end{aligned}$$

(Single Block SDP Problem)

Remark. 1) It is evident that, as long as the distinguishability assumption holds for all lines (*i.e.*, the distance from an inlier line to its corresponding vanishing point is smaller than the maximum inlier threshold, while the distance to other vanishing points is greater than the maximum inlier threshold), [Single Block SDP Problem](#) will consistently identify the vanishing point with the largest inlier set. 2) In Supplementary Material Sec. 1.3, we provide a proof of tight relaxation (*i.e.*, zero duality gap) under noise-free and outlier-free conditions.

4.3.2. Manhattan Post-Refinement

After three iterations, given three coarse estimated vanishing points $\{\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3\}$ and the corresponding set of line labels $\{\mathbf{l}_1, \mathbf{l}_2, \mathbf{l}_3, \mathbf{l}_{\text{outlier}}\}$, we perform [Manhattan Post-Refinement](#) to obtain the refined vanishing points $\{\mathbf{d}_1^*, \mathbf{d}_2^*, \mathbf{d}_3^*\}$:

$$\begin{aligned} \min_{\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3} \quad & \sum_{i=1}^3 \sum_{j \in \mathbf{l}_i} (\mathbf{d}_i^\top \mathbf{n}_j)^2 \\ \text{s.t.} \quad & \mathbf{d}_1^\top \mathbf{d}_1 = 1, \mathbf{d}_2^\top \mathbf{d}_2 = 1, \mathbf{d}_3^\top \mathbf{d}_3 = 1, \\ & \mathbf{d}_1^\top \mathbf{d}_2 = 0, \mathbf{d}_2^\top \mathbf{d}_3 = 0, \mathbf{d}_1^\top \mathbf{d}_3 = 0. \end{aligned}$$

(Manhattan Post-Refinement)

Remark. 1) Although **Single Block SDP Problem** does not rely on the Manhattan world assumption, it still yields the optimal inlier line set. When we incorporate the Manhattan world prior into **Manhattan Post-Refinement**, the final refined vanishing points retain the optimality as in **Full SDP Problem**. 2) It is well known that the optimization time for SDP methods grows cubically with the problem size, making them inefficient for large-scale problems. To overcome this issue, we adopt the strategy from [18], which randomly samples a fixed number of lines for each **Single Block SDP Problem**. While this aspect is not the main contribution of our work, it provides a highly advantageous trade-off for our algorithm in large-scale settings, as shown in Fig. 5.

5. Experiments

5.1. Comparisons against Traditional Methods

5.1.1. Experimental Settings

Baseline methods. We compare our method, **GlobustVP**, with the following state-of-the-art traditional approaches:

- RANSAC-based method [42] (denoted as **RANSAC**);
- J-Linkage-based method [38] (denoted as **J-Linkage**);
- T-Linkage-based method [30] (denoted as **T-Linkage**);
- BnB-based method [6] (denoted as **BnB**);
- RANSAC-BnB hybrid method [25] (denoted as **Quasi-VP**).

Implementation details. Our approach is implemented in MATLAB, using MOSEK [3] as the SDP solver. All experiments are conducted on a laptop with an Intel Core i5-8250U CPU (1.80 GHz) and 8GB of RAM.

Evaluation metrics. We follow [25, 28] to evaluate the accuracy of the image line association in terms of precision $\uparrow$, recall $\uparrow$, and the F_1 -score $\uparrow$. Specifically, precision is computed as $N_c/(N_c + N_w)$ where N_c and N_w denote the counts of correctly and incorrectly identified inliers, respectively. Recall is defined as $N_c/(N_c + N_m)$ where N_m represents the number of missing inliers. The F_1 -score, which considers both precision and recall, is given by $(2 \times \text{precision} \times \text{recall})/(\text{precision} + \text{recall})$. For the evaluation of the real-world dataset, we use the root mean square of the consistency error $\downarrow$ [25, 37, 42]. This metric is particularly suitable for 3D contexts, as image uncertainties typically arise in the image domain [42]. Specifically, the consistency error quantifies the distance from an endpoint of the image line l to the virtual line v , where v is defined by the estimated vanishing point and the midpoint of the line l associated with this vanishing point.

Evaluation setup. For the synthetic dataset, we generate three distinct sets of parallel 3D lines, ensuring that each pair of lines from different sets is mutually orthogonal. Using a virtual camera, we project these lines onto the virtual

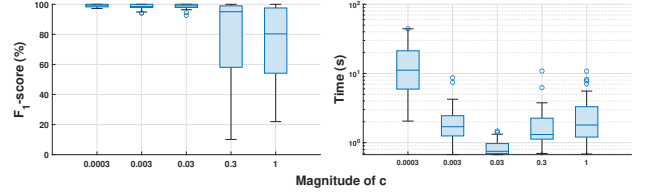


Figure 3. Effect of different values of the maximum inlier threshold c . Best viewed in high resolution.

image plane, resulting in three sets of image lines. This virtual image has a resolution of 640×480 pixels and a focal length of 800 pixels, with the principal point located at its center. We then introduce perturbations to the endpoints of these image lines using zero-mean Gaussian noise with a standard deviation of σ pixels. Additionally, to generate outlier image lines, we replace a portion of the endpoints with random pixels within the virtual image. The number of lines associated with each vanishing point is randomly distributed, with each vanishing point having a respective number of lines in our experiments. It is important to note that this distribution is not uniform across vanishing points. All statistical analyses are performed across 100 independent Monte Carlo trials, unless otherwise specified.

5.1.2. Synthetic Dataset

Effect of c . To better understand **GlobustVP**, we explore different values of c (*i.e.*, maximum inlier threshold introduced in 4.1), ranging from 0.0003 to 1. We fix the number of lines at 120, the noise level σ at 1 pixel, and the outlier ratio at 30%. The results in Fig. 3 show that the F_1 -score decreases as c increases because larger values of c cause more outliers to be misclassified as inliers, thus reducing the accuracy. Furthermore, the runtime decreases as c increases, reaching a minimum at $c = 0.03$, after which it increases again. Therefore, we choose $c = 0.03$ for all subsequent experiments, as it provides a good balance between accuracy and computational efficiency.

Accuracy comparisons across outlier ratios. We use 60 lines, set the noise level at $\sigma = 3$, and increase the outlier ratio from 0% (all inliers) to 70%. Fig. 6 presents F_1 -score across different outlier ratios. **RANSAC**, **J-Linkage**, and **T-Linkage** maintain robustness only at low outlier ratios, but degrade significantly when outliers exceed 20%, while **Quasi-VP** starts to break down at 40%. **BnB** generally handles high outlier ratios well, while occasional convergence issues reduce its accuracy. In contrast, **GlobustVP** consistently demonstrates superior robustness and accuracy across varying outlier ratios. The complete results, including precision and recall, are provided in Supplementary Material Sec. 3.

Sampling acceleration in large-scale scenario. To evaluate sampling acceleration in large-scale scenarios, we generate 120 lines, setting the noise level at $\sigma = 3$ and the out-

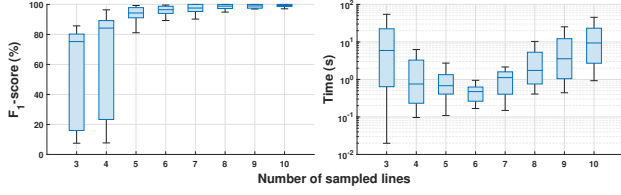


Figure 4. Accuracy and efficiency comparisons with respect to the number of sampled lines on the synthetic dataset. Best viewed in high resolution.

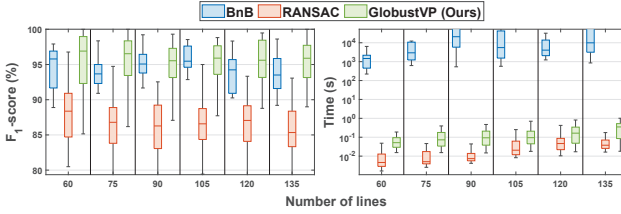


Figure 5. Accuracy and efficiency comparisons of **RANSAC** [42] and **BnB** [6] on the synthetic dataset with respect to the number of image lines. Complete comparisons of all baseline methods are provided in Supplementary Material Sec. 3. Best viewed in color and high resolution.

lier ratio at 20%. We vary the number of sampled lines from 3 to 10, with an interval of 1. Fig. 4 shows both the F_1 -score and runtime. The total runtime decreases as the number of sampled lines increases, reaching a minimum at 6, and then increases again. In contrast, the accuracy (*i.e.*, F_1 -score) improves as the number of sampled lines grows, stabilizing at 6 sampled lines and remaining consistent for larger values. Therefore, in the following efficiency experiments, we fix the number of sampled lines at 6, which provides the optimal trade-off in our method.

Accuracy and efficiency comparisons across line counts. Using the sampling acceleration strategy described above, where the optimal number of sampled lines is determined to be 6, we conduct a thorough assessment of accuracy and efficiency with respect to the number of image lines. Specifically, we set the noise level at $\sigma = 3$ and the outlier ratio at 20%, while varying the number of image lines from 60 to 135. For clarity, we present comparisons here specifically with **RANSAC** and **BnB**, while complete results of all the baseline methods are provided in Supplementary Material Sec. 3. As shown in Fig. 5, we report both the F_1 -score and the runtime as the number of image lines increases. While **RANSAC** achieves the highest efficiency, it suffers from a lower accuracy (*i.e.*, F_1 -score). In contrast, **GlobustVP** achieves significantly faster runtime compared to **BnB** while maintaining robust accuracy. This balance between efficiency and accuracy highlights the practical applicability of our method in real-world scenarios.

Datasets	YUD [10]			SU3 [45]		
Metrics	AA@3° ↑	AA@5° ↑	AA@10° ↑	AA@3° ↑	AA@5° ↑	AA@10° ↑
J-Linkage [38]	57.7	69.3	80.5	70.2	77.6	83.8
Contrario-VP [34]	58.3	71.1	80.7	66.8	74.0	81.6
Quasi-VP [25]	57.8	72.5	84.3	72.2	78.8	81.8
Zhai <i>et al.</i> [41]	48.9	63.1	76.5	65.4	75.4	82.2
NeurVPS [44]	52.2	64.2	78.1	93.2	95.2	97.8
CONSAC [17]	61.9	73.5	83.4	79.2	85.8	90.6
Lin <i>et al.</i> [27]	63.3	75.6	85.7	84.2	90.4	95.0
PARSAC [16]	64.7	77.2	85.6	86.2	90.6	95.4
GlobustVP (Ours)	67.6	87.3	96.1	80.2	86.8	92.4

Table 1. Angular accuracy on YUD [10] and SU3 [45] datasets. **GlobustVP** achieves the best results on YUD [10], and is competitive on SU3 [45]. The **best** and **second-best** performance for each metric are highlighted.

5.1.3. Real-World Dataset

To evaluate different methods on real-world images, we conduct experiments using the York Urban Database (YUD) [10]. This dataset is widely used for evaluating the estimation of vanishing points and the association of lines in the Manhattan world. It consists of 102 calibrated images, each with a resolution of 640×480 pixels, taken from both indoor and outdoor environments. Each image contains a set of manually annotated inlier line segments (all inliers), corresponding to either 2 or 3 vanishing points, along with the ground truth line-VP associations. Figs. 1 and 7 show line-VP association results obtained by **GlobustVP** and other state-of-the-art techniques. A detailed analysis of F_1 -score and consistency error on all images of YUD [10], along with further quantitative comparisons, is provided in Supplementary Material Sec. 5.1.

5.2. Comparisons against Learning-Based Methods

While our method outperforms traditional approaches on both synthetic and real data, it is worth noting that learning-based methods have recently demonstrated promising results in vanishing point estimation.

5.2.1. Experimental Settings

Datasets. In addition to YUD [10], we also evaluate on the SceneCity Urban 3D (SU3) wireframe dataset [45]. The SU3 dataset consists of 23K synthetic images, divided into 80% for training, 10% for validation, and 10% for testing.

Baseline methods. We compare our method, **GlobustVP** with the state-of-the-art learning-based methods and traditional, including [41], **NeurVPS** [44], **CONSAC** [17], [27], and **PARSAC** [16] on YUD [10] and SU3 [45] datasets.

Evaluation protocol. Following [27], we train [41] and **CONSAC** from scratch on SU3 [45], while using official code and pre-trained models for **NeurVPS**, [27], and **PARSAC**, which are typically well-trained on SU3 [45]. Additionally, we evaluate traditional methods, including **J-Linkage**, **Contrario-VP**[34], and **Quasi-VP**. **Contrario-VP** employs an a-contrario framework to constrain vanishing points on the horizon line. To ensure consistency,

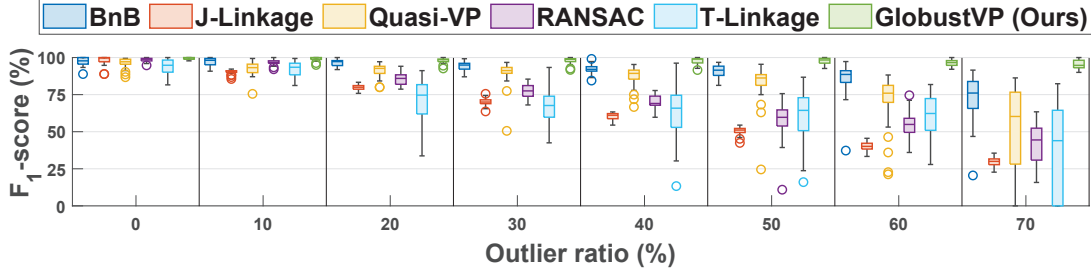


Figure 6. Accuracy comparison on the synthetic dataset with respect to the outlier ratios: boxplot of F_1 -score. Results for precision and recall are provided in Supplementary Material Sec. 3. Best viewed in color and high resolution.

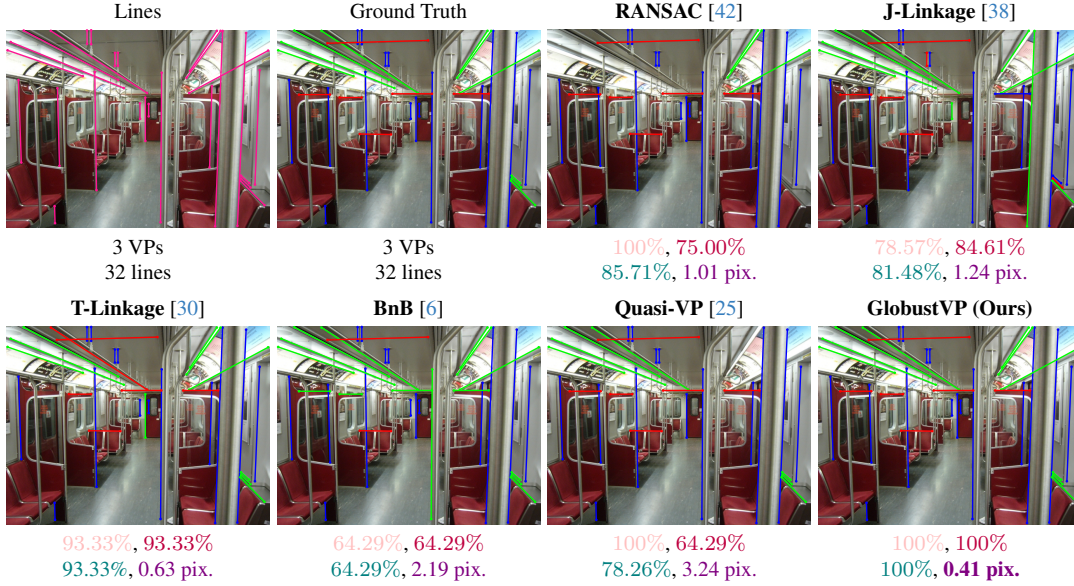


Figure 7. Representative comparisons on YUD [10] using the manually extracted image lines. Different line-VP associations are shown in respective colors. The numbers below each image represent the respective precision $\uparrow$, recall $\uparrow$, F_1 -score $\uparrow$, and consistency error $\downarrow$ of line-VP association. Best viewed in color and high resolution.

we follow [44] and adopt the angle accuracy (AA) metric, which measures the angle between the estimated and ground truth dominant directions. We then calculate the percentage of estimates with angular differences below pre-defined thresholds (3° , 5° , and 10°).

5.2.2. Results

As shown in Tab. 1, **GlobustVP** achieves the highest AA scores on YUD [10] across all thresholds, attaining an AA@ 10° of 96.1%, which significantly outperforms both learning-based and traditional approaches. On SU3 [45], **GlobustVP** demonstrates competitive performance, particularly compared to traditional baselines, while learning-based methods, such as **NeurVPS** and **PARSAC**, achieve better results due to their training on similar data. This difference in generalization capability highlights the robustness of **GlobustVP** in diverse settings, as it does not rely on data-specific training. By effectively balancing accuracy, robustness, and generalization, **GlobustVP** proves

well-suited for real-world applications where data-driven methods often struggle with limited generalization.

6. Conclusions

In this paper, we introduce convex relaxation techniques to VP estimation for the first time. Specifically, we propose a “soft” association scheme that allows for jointly inferring the line-VP association and locating each VP. The core for establishing such a scheme is the proposed truncated multi-selection error, an attribute to which the original primal problem can be reformulated to a QCQP form and further relaxed to a convex SDP problem. Besides, we propose an iterative solver, **GlobustVP**, for efficiently solving the resulting SDP problem, enjoying efficient computation while preserving global optimality under mild conditions. Experimental results justify the effectiveness of our algorithm, outperforming existing methods in terms of efficiency, robustness to outliers, and guaranteed global optimality.

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